Mohamed Yassine Ben Romdhane

AI & FULL-STACK ENGINEER

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SUMMARY

AI and full-stack engineer (ESPRIM, 2026) building end-to-end RAG systems and web/mobile platforms. Designed a source-grounded documentation assistant at Capgemini Engineering with FastAPI, pgvector, and multi-provider LLM routing. Builds React and React Native frontends backed by Node.js, Python, and PostgreSQL.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL
Frontend: React, React Native, Next.js, Angular, Tailwind CSS, Expo
Backend: Node.js, Express, Fastify, FastAPI, Spring Boot, REST APIs
AI & Data: RAG, LLMs, embeddings, pgvector, hybrid retrieval, tool calling, Gemini, NVIDIA NIM, scikit-learn
Databases: PostgreSQL, MongoDB, Redis, Supabase
Tools: Docker, Docker Compose, Git, Socket.IO, MQTT

WORK EXPERIENCE

AI Engineer Intern — Capgemini Engineering — Feb 2026 – Jun 2026
Final-year project · Source-grounded documentation assistant (RAG)

- Built a Retrieval-Augmented Generation (RAG) assistant that answers natural-language questions across 4 document formats — PDF, DOCX, Markdown, and wiki exports — citing the exact sources behind every answer.
- Designed a backend split into 2 services — Node.js/Express for authentication and conversation orchestration, Python/FastAPI for parsing, chunking, embeddings, and retrieval — with PostgreSQL + pgvector as the shared store.
- Implemented hybrid dense-vector and full-text retrieval with Reciprocal Rank Fusion (RRF), plus a verification pass that rejects unsupported answers.
- Added task-specific LLM routing across 2 providers (NVIDIA NIM for generation and tool calls, Gemini for summaries), query-rewrite retries, and stream-level error handling.
- Validated 11 end-to-end scenarios — access control, source attribution, provider failover, and streaming — all passing, with a median chat latency of 1.9 s on an 18-document corpus in local tests.

Full-Stack & Mobile Engineer — YBInnovations — 2025
Project · Wave, a campus community platform for students and associations

- Built a cross-platform Expo/React Native app with 6 modules: events, carpooling, housing, tutoring, deals, and real-time chat.
- Engineered a TypeScript/Express API over PostgreSQL and MongoDB with JWT authentication and per-user data scoping.
- Built a Redis-backed push notification pipeline and integrated Lydia payments with webhook handling.
- Implemented Socket.IO real-time messaging and state management across the 6 modules.

Full-Stack Developer & Data Engineer — MyMatch — 2024
Project · Football match booking and management platform

- Developed REST APIs for 3 core workflows: authentication, bookings, and match management.
- Implemented match recommendations from player and match data, with query optimization at the database level.
- Built booking and match-management frontend modules; contributed to schema design and query tuning.

EDUCATION

Engineering Degree in Computer Science — Data Science track (2023 – 2026)
ESPRIM — École Supérieure Privée d'Ingénieurs de Monastir
National engineering degree; final-year project at Capgemini Engineering.

LANGUAGES

Arabic (Native) · French (Fluent) · English (Fluent)